WORKED SOLUTION

STEP I 2001, Question 9

Mechanics • Interception and Relative Velocity

STEP I 2001, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s1-q9

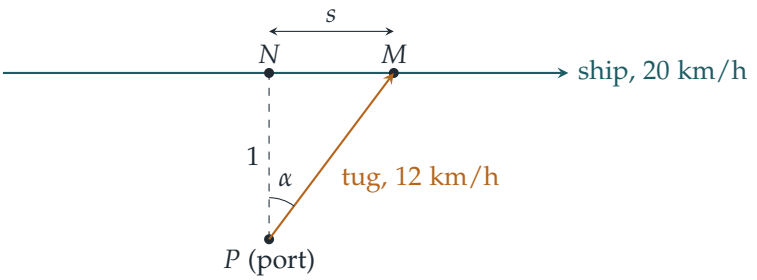
The key idea

Leaving as late as possible means the tug must go flat out in a straight line, so the only real choice is *where* on the ship's track to meet it. For each possible meeting point, work out the latest time the tug could leave and still be there in time. Then maximise that departure time over the meeting point. It's a one-variable optimisation once the set-up is right.

Solution

Setting up

Treat the ship and the tug as particles. Put the port P at the origin, and let the ship travel along the line $y = 1$ in the positive x direction (distances in km, times in hours). The foot of the perpendicular from P to the track is $N = (0, 1)$, the ship's closest point to the port.



Suppose the meeting takes place at the point $M = (s, 1)$. The ship's position along the track is a linear function of time, so it reaches M at time

$$t_M = \frac{s - c}{20},$$

where c is a constant fixed by where the ship starts. (The ship is still approaching, so c is less than any meeting position we will need.)

> **Ask yourself** Why must the tug travel at full speed in a straight line?

If the tug is at M at time t_M having left at time T, it has covered at least the straight-line distance $PM = \sqrt{s^2+1}$ at no more than 12 km/h, so $t_M - T \geq \sqrt{s^2+1}/12$. Equality is attained by heading straight for M at top speed. Anything slower or less direct forces an earlier departure.

So the latest departure time that allows a meeting at M is

$$T(s) = \frac{s-c}{20} - \frac{\sqrt{s^2+1}}{12},$$

and we want the largest value of $T(s)$ over all s.

Maximising the departure time

Differentiating,

$$T'(s) = \frac{1}{20} - \frac{s}{12\sqrt{s^2+1}}.$$

Setting $T'(s) = 0$ gives

$$\frac{s}{\sqrt{s^2+1}} = \frac{12}{20} = \frac{3}{5}.$$

This needs $s > 0$. Squaring, $25s^2 = 9(s^2+1)$, so $s^2 = \frac{9}{16}$ and $s = \frac{3}{4}$.

This is a maximum, and a global one: $\sqrt{s^2+1}$ is convex, so T (a linear function minus a convex one) is concave. Equivalently, $\dfrac{s}{\sqrt{s^2+1}}$ is increasing in s, so $T'(s) > 0$ for $s < \frac{3}{4}$ and $T'(s) < 0$ for $s > \frac{3}{4}$. Note that the constant c has dropped out: the meeting point doesn't depend on where the ship started.

The tug therefore meets the ship $\frac{3}{4}$ km beyond N, and the distance it travels is

$$PM = \sqrt{\tfrac{9}{16}+1} = \sqrt{\tfrac{25}{16}} \quad\Longrightarrow\quad \boxed{PM = \tfrac{5}{4}\text{ km} = 1.25\text{ km}.}$$

The tug is at sea for $\frac{5/4}{12} = \frac{5}{48}$ hours, which is $6\frac{1}{4}$ minutes.

A geometric reading

The tug's heading makes angle α with PN, where $\sin\alpha = \frac{s}{\sqrt{s^2+1}} = \frac{3}{5}$, so $\sin\alpha$ is the ratio of the speeds, $12/20$. The tug's velocity is $(7.2, 9.6)$, and its velocity relative to the ship is $(7.2-20, 9.6) = (-12.8, 9.6)$. Their dot product is $-92.16 + 92.16 = 0$. At the critical departure time, the tug's velocity is perpendicular to its velocity relative to the ship. This is the standard condition for a slower pursuer that only just manages an interception.

Checks

- A brute-force search over a fine grid of meeting points, for several different starting positions of the ship, always puts the maximum of $T(s)$ at $s = 0.75$, so the tug's distance is 1.25 km.
- Stepping the motion numerically from the latest departure time with heading $(\frac{3}{5}, \frac{4}{5})$, the tug reaches the ship. Leaving just 0.002 hours later, no heading gets within 20 m of the ship.
- Sanity check: if the tug were as fast as the ship, $\sin\alpha$ would be 1, and the meeting point would run off to infinity. That makes sense, since a tug of equal speed can chase the ship indefinitely.

Common slips

- Meeting the ship at its closest point N (tug distance 1 km). That minimises the tug's *distance*, not its departure time: by going a little further forward, the tug gains time because the ship still has to sail there.
- Minimising the tug's travel time $\sqrt{s^2+1}/12$ without the $s/20$ term. The quantity to maximise is the departure time, which involves both boats.

Beyond STEP

Maximising a difference of arrival times is a small instance of an optimal-control or pursuit problem. The ratio-of-speeds condition $\sin\alpha = 12/20$ is the same geometry as the Mach cone behind a supersonic aircraft: the sine of the cone's half-angle is the ratio of the sound speed to the aircraft's speed, and the slow tug here plays the part of sound chasing a fast source. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats relative velocity and frames carefully, and Siklos, *Advanced Problems in Mathematics*, has more STEP-style optimisation in mechanics.

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